Rules: Soft and Strong Governance

The Role of Governance Rules

The Role of Governance Rules I

Research software projects make dozens of implicit governance decisions: what test-coverage threshold is acceptable, how manuscript sections should be ordered, which citation fields are mandatory. Left implicit, these decisions drift silently across projects in a monorepo, eroding the consistency that makes the repository valuable as a public exemplar. The rules layer makes governance explicit, versioned, and machine-enforceable.

A **rule set** in the template repository is a directory under `rules/<scope>/<name>/` containing a typed manifest (`rules.yaml`) and two subdirectories of rule files:

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```
<name>/
   rules.yaml          - manifest (type, scope, version,
rule_kinds)
   soft/               - Markdown guideline files
(human-readable, prompt-like)
   strong/             - YAML constraint schemas
(machine-enforceable)
```

This two-tier architecture reflects the distinction between *guidance* (which humans follow approximately) and *constraints* (which pipelines enforce precisely) — a distinction also recognised in enterprise application architecture (Fowler 2002). fig. 3 shows both template rule sets split into their soft and strong branches: each branch is independently discoverable, so a consumer that only cares about machine-enforceable constraints never has to parse guideline prose, and vice versa.

Soft Rules: Style and Process Guidelines

Soft Rules: Style and Process Guidelines I

Soft rules are Markdown files in `soft/`. They encode preferences and conventions that cannot easily be expressed as boolean constraints but that human reviewers and AI agents can apply contextually. Examples include:

Soft Rules: Style and Process Guidelines I

- **Style preferences**: "Prefer active voice in manuscript sections." "Use `\module{}` macros for all code identifiers."
- **Process guidelines**: "Tag pull requests with a review-stage label before requesting review." "Update `TODO.md` before closing an issue."
- **Citation conventions**: "Cite primary sources rather than textbooks where possible."

Soft Rules: Style and Process Guidelines I

Soft rules are treated as guidance: deviations surface as suggestions in code review and manuscript audit reports, not as pipeline blockers. This makes the soft layer suitable for evolving preferences that should not break automated checks.

Strong Rules: Hard Constraints

Strong Rules: Hard Constraints I

Strong rules are YAML files in `strong/`. Each file defines one named constraint:

```yaml
rule:
  name: coverage-gate
  kind: strong
  description: "Minimum test coverage threshold for
src/ modules."
  applies_to: "projects/*/src/"
  enforcement: fail_on_violation
  constraints:
    infrastructure: 60
    project_src: 90
    public_api: 95
```

Strong Rules: Hard Constraints I

The `enforcement: fail_on_violation` field signals that a pipeline must halt and report when this rule is violated. Strong rules are suitable for invariants that, if broken, indicate a genuine defect rather than a style preference: coverage below the source-declared threshold means tests are missing; a manuscript section without an abstract means the document is incomplete.

Strong Rules: Hard Constraints I



Strong Rules: Hard Constraints I

Figure 3: Rule hierarchy: the two template rule sets, each split into a machine-enforceable `strong/` branch and a guidance-only `soft/` branch.

The Two Template Rule Sets

`template_project_rules` |

This rule set governs software projects throughout the template repository. Its strong rules currently comprise:

template_project_rules

File	Constraint
strong/coverage-gate.yaml	Minimum coverage: infrastructure 60%, project src/ 90%, public API 95%
strong/module-structure.yaml	Required directory layout: src/, tests/, scripts/, manuscript/

Its soft rules provide guidance on code style, commit message conventions, and pull-request labelling.

This rule set governs research manuscripts. Its strong rules comprise:

template_manuscript_rules

File	Constraint
strong/reference-schema.yaml	Required BibTeX fields and cite-key format constraints
strong/section-schema.yaml	Required manuscript sections, ordering, and minimum word counts

In the current pipeline run, **2 of 2 rule sets** validated successfully (fig. 5).

Rule Application Module

Rule Application Module I

The src/template_pools_rules_tools/rules/rules_applier.py module exposes three functions:

```
from template_pools_rules_tools.rules.rules_applier
import (
    load_soft_rules,
    load_strong_rules,
    validate_against_rules,
)


soft   = load_soft_rules("template_project_rules")
# list[dict]
strong = load_strong_rules("template_project_rules")
# list[dict]
result =
validate_against_rules("template_project_rules")
# → {"status": "ok" | "partial" | "missing",
"soft_count": N, "strong_count": N}
```

`validate_against_rules()` performs two checks: (1) the `rules.yaml` manifest is parseable YAML; (2) every rule file in `soft/` and `strong/` is parseable YAML. It returns a status of `"ok"` when both checks pass, `"partial"` when the manifest exists but some rule files are missing or malformed, and `"missing"` when the rule set directory is absent entirely. This graduated status enables the integration pipeline to distinguish between a rule set that has not yet been created (acceptable during active development) and one that is present but broken (actionable defect).

Rules and Manuscript Variables

Rules and Manuscript Variables I

Strong rule validation counts are injected into the manuscript through the token system. The token 2 expands to the count of rule sets that returned `status="ok"` during the integration run. This creates a verifiable link between the pipeline's actual behaviour and the manuscript's claims — the manuscript cannot assert successful validation without the pipeline having actually succeeded.

Beyond Structural Validation: Semantic Rule Evaluation

Beyond Structural Validation: Semantic Rule Evaluation I

`validate_against_rules()` (described above) performs *structural* validation only: it confirms that `rules.yaml` and every file in `soft//strong/` parse as YAML. It does not check whether the constraints those strong-rule files declare are actually satisfied by the current project. That semantic layer lives in a separate module, `src/template_pools_rules_tools/rules/strong_rule_evaluator.py`, exposed via `scripts/04_validate_strong_rules.py`:

Beyond Structural Validation: Semantic Rule Evaluation I

```
from template_pools_rules_tools.rules.strong_rule_eva
luator import evaluate_strong_rules,
load_rule_context_from_project

context =
load_rule_context_from_project(project_root)
result =
evaluate_strong_rules("template_project_rules",
context)
# → {"rule_set": ..., "evaluations": [...], "passed":
bool, "violation_count": int}
```

Beyond Structural Validation: Semantic Rule Evaluation I

`evaluate_strong_rules()` dispatches each strong-rule YAML file to a rule-kind-specific evaluator function keyed by its declared kind, via a small dispatch table covering all four strong-rule kinds that exist across both rule sets:

Beyond Structural Validation: Semantic Rule Evaluation I

Kind	Evaluator	Checks
`coverage_threshold`	`_evaluate_coverage_threshold`	Measured coverage percentages (from `context["coverage"]`) against each constraint's declared `minimum_line_coverag e`
`module_structure`	`_evaluate_module_structure`	Required project directory layout (`src/`, `tests/`, `scripts/`, `manuscript/`) actually exists
`section_schema`	`_evaluate_section_schema`	Required manuscript sections, ordering, and forbidden placeholder headings (`TODO`, `Draft`, etc.)
`reference_schema`	`_evaluate_reference_schema`	Required BibTeX fields and cite-key format constraints on every parsed reference entry

Beyond Structural Validation: Semantic Rule Evaluation I

Each evaluator distinguishes structured violation *reasons* rather than collapsing everything to a boolean — for example, `coverage_threshold` reports separately whether a key was absent from context (a context-completeness issue), non-numeric (a context-shape issue), or numeric-but-below-minimum (a genuine rule violation). This is what lets the pipeline tell a maintainer *why* a rule failed, not merely *that* it failed — directly addressing the "actionable defect" distinction introduced in the previous section.

Beyond Structural Validation: Semantic Rule Evaluation I

Crucially, `section_schema` and `reference_schema` are not evaluated against synthetic fixtures — `load_rule_context_from_project()` (in `scripts/04_validate_strong_rules.py`) builds their context by parsing *this project's own, current* `manuscript/references.bib` into structured reference entries and extracting the real #-level headings from every file under `manuscript/*.md`. Running `uv run python projects/templates/template_pools_rules_tools/scripts/04_validate_strong_rules.py` therefore semantically validates this exact manuscript's own bibliography and section structure, live, on every invocation — re-run that command to see the current evaluation and violation counts rather than trusting a number printed here, since either count can legitimately change as the manuscript grows.